MAT 3141 Assignment (Applied Statistics & Probability)

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from statistics import mean, median, mode
from scipy.stats import gmean, hmean
```

Question 1

a) Create student dataset

In [2]:
```
data = {
    "ID": [241311056, 241311057, 241311058, 241311059, 241311060],
    "Name": ["Md. Oahadur Rahman Evu", "Yalmai Lumia ", "Abdullah Al Mahfuz",
    "Gender": ["Male", "Female", "Male", "Female", "Female"],
    "Home District": ["Naogaon", "Rajshahi", "Rajshahi", "Naogaon", "Chapainaw
    "Age": [22, 22, 23, 21, 23],
    "Parents Income": [45000, 25000, 7000, 15000, 20000],
    "CGPA": [3.72, 3.46, 3.41, 3.17, 3.50]
}
```

b) Print individual results

In [3]:
```
print("Individual Student Information:\n")

for i in range(len(data["ID"])):
    print("----")
    print(f"ID: {data['ID'][i]} \t| Name: {data['Name'][i]}")
    print(f"\t\t| Gender: {data['Gender'][i]}")
    print(f"\t\t| Home District: {data['Home District'][i]}")
    print(f"\t\t| Age: {data['Age'][i]}")
    print(f"\t\t| Parents Income: {data['Parents Income'][i]}")
    print(f"\t\t| CGPA: {data['CGPA'][i]}")
```

```
Individual Student Information:

----
ID: 241311056         | Name: Md. Oahadur Rahman Evu
                | Gender: Male
                | Home District: Naogaon
                | Age: 22
                | Parents Income: 45000
                | CGPA: 3.72
----
ID: 241311057         | Name: Yalmai Lumia
                | Gender: Female
                | Home District: Rajshahi
                | Age: 22
                | Parents Income: 25000
                | CGPA: 3.46
----
ID: 241311058         | Name: Abdullah Al Mahfuz
                | Gender: Male
                | Home District: Rajshahi
                | Age: 23
                | Parents Income: 7000
                | CGPA: 3.41
----
ID: 241311059         | Name: Nishat Tabassum Promi
                | Gender: Female
                | Home District: Naogaon
                | Age: 21
                | Parents Income: 15000
                | CGPA: 3.17
----
ID: 241311060         | Name: Mst.Farhana Jahan
                | Gender: Female
                | Home District: Chapainawabganj
                | Age: 23
                | Parents Income: 20000
                | CGPA: 3.5
```

c) Read using DataFrame

In [4]:

```
df = pd.DataFrame(data)

print("\nCombined DataFrame:\n")
print(df.to_string(index=False))
```

```
Combined DataFrame:

       ID                   Name Gender   Home District  Age  Parents Income  CGPA
241311056 Md. Oahadur Rahman Evu   Male         Naogaon   22           45000  3.72
241311057          Yalmai Lumia  Female        Rajshahi   22           25000  3.46
241311058     Abdullah Al Mahfuz   Male        Rajshahi   23            7000  3.41
241311059 Nishat Tabassum Promi  Female         Naogaon   21           15000  3.17
241311060      Mst.Farhana Jahan Female Chapainawabganj   23           20000  3.50
```

d) AM, GM, HM, Median, Mode

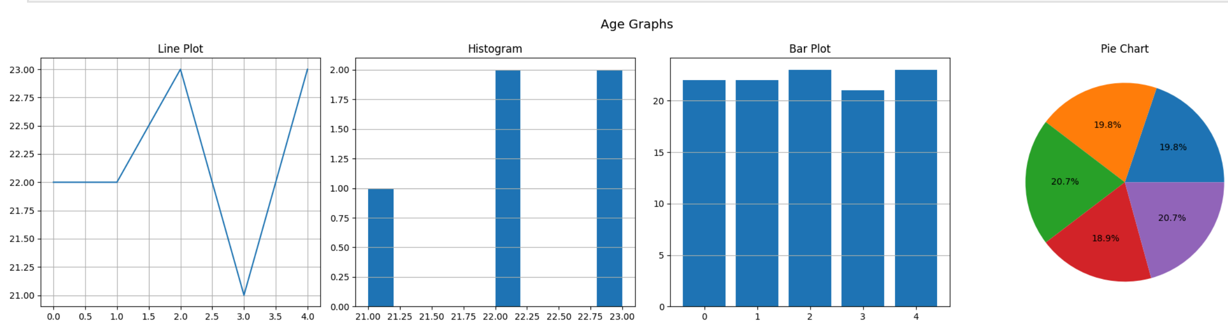
In [5]:

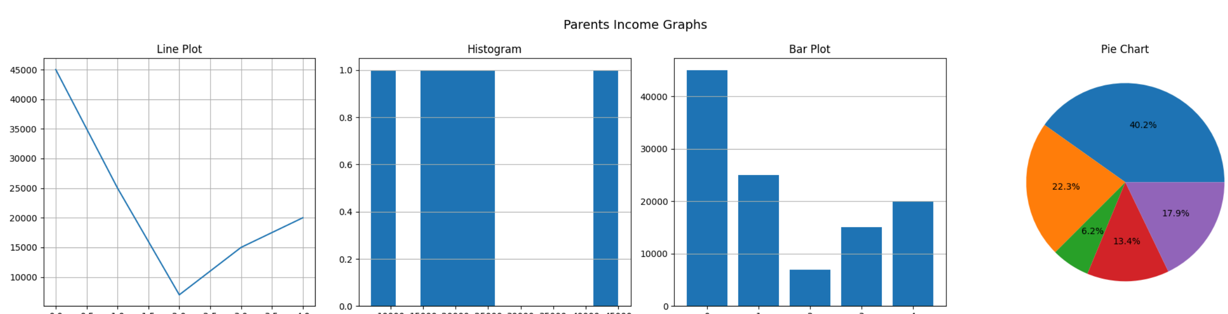
```python
columns = ['Age','Parents Income','CGPA']

def calculate_stats(column, name):
    print(f"\nStatistics for {name}:")
    print("Arithmetic Mean:", mean(column))
    print("Geometric Mean:", gmean(column))
    print("Harmonic Mean:", hmean(column))
    print("Median:", median(column))
    print("Mode:", mode(column))

calculate_stats(data["Age"], "Age")
calculate_stats(data["Parents Income"], "Parents Income")
calculate_stats(data["CGPA"], "CGPA")
```

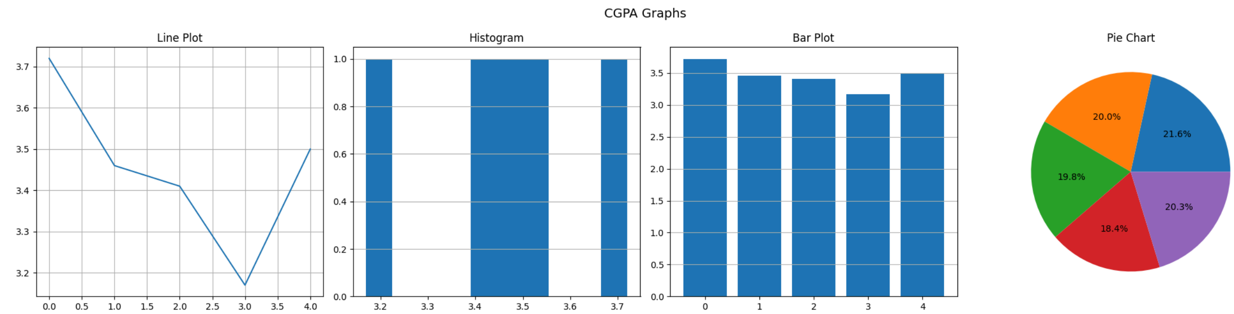
```
Statistics for Age:
Arithmetic Mean: 22.2
Geometric Mean: 22.18728007912323
Harmonic Mean: 22.174457429048417
Median: 22
Mode: 22

Statistics for Parents Income:
Arithmetic Mean: 22400
Geometric Mean: 18822.372476636763
Harmonic Mean: 15540.2072027627
Median: 20000
Mode: 45000

Statistics for CGPA:
Arithmetic Mean: 3.452
Geometric Mean: 3.4474672155396506
Harmonic Mean: 3.442906409067063
Median: 3.46
Mode: 3.72
```

e) Plots

In [6]:
```python
def draw_graphs(column, title):

    plt.figure(figsize=(20, 5))

    # Line Plot
    plt.subplot(1, 4, 1)
    plt.plot(column)
    plt.title("Line Plot")
    plt.grid()

    # Histogram
    plt.subplot(1, 4, 2)
    plt.hist(column)
    plt.title("Histogram")
    plt.grid(axis='y')

    # Bar Plot
    plt.subplot(1, 4, 3)
    plt.bar(range(len(column)), column)
    plt.title("Bar Plot")
    plt.grid(axis='y')

    # Pie Chart
    plt.subplot(1, 4, 4)
    plt.pie(column, autopct='%1.1f%%')
    plt.title("Pie Chart")

    plt.suptitle(f"{title} Graphs", fontsize=14)
    plt.tight_layout()
    plt.show()

draw_graphs(data["Age"], "Age")
draw_graphs(data["Parents Income"], "Parents Income")
draw_graphs(data["CGPA"], "CGPA")
```

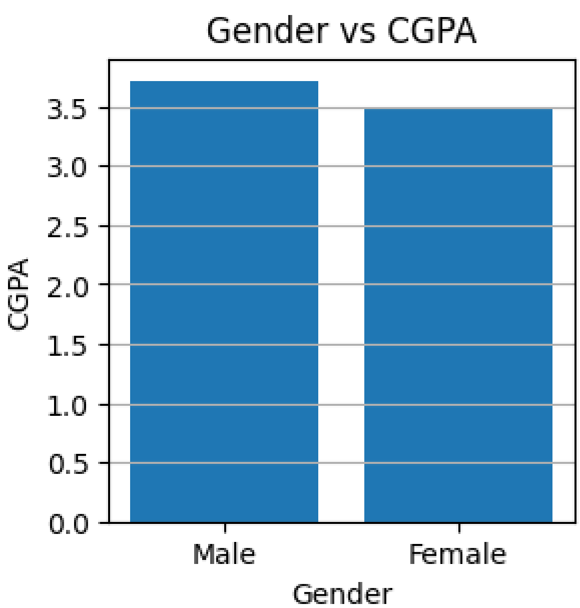






f) Gender vs CGPA

In [7]:
```python
plt.figure(figsize=(3, 3))
plt.bar(data["Gender"], data["CGPA"])
plt.xlabel("Gender")
plt.ylabel("CGPA")
plt.title("Gender vs CGPA")
plt.grid(axis='y')
plt.show()
```



Question 2

a) Input variable X

In [8]:
```python
X = [
11,17,15,22,11,16,19,24,29,25,
18,18,25,26,32,14,17,20,23,27,
30,12,24,36,18,15,20,28,33,38,
34,13,12,16,20,22,29,19,23,31
]
```

b) AM, GM, HM, Median, Mode

In [9]:
```
print("Arithmetic Mean:", mean(X))
print("Geometric Mean:", gmean(X))
print("Harmonic Mean:", hmean(X))
print("Median:", median(X))
print("Mode:", mode(X))
```

```
Arithmetic Mean: 22.05
Geometric Mean: 20.88281402828289
Harmonic Mean: 19.73579885307883
Median: 21.0
Mode: 18
```

c) Frequency Distribution

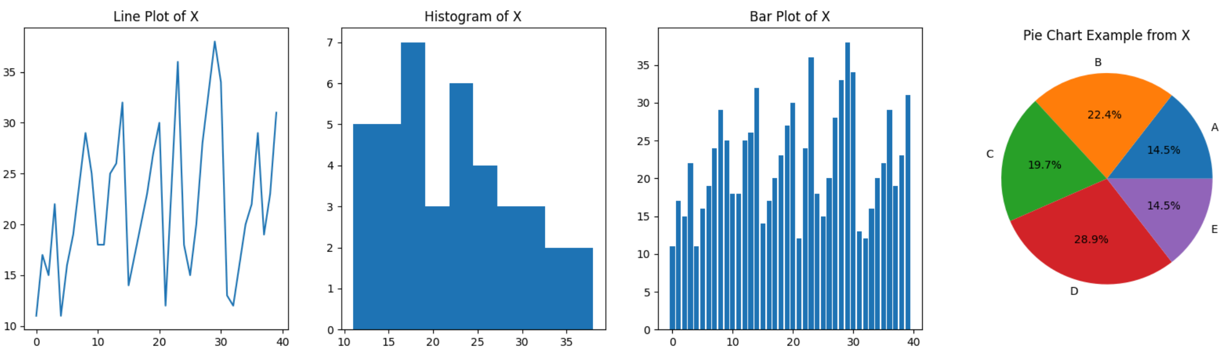
In [10]:
```
# Number of class intervals (Sturges' Rule)
k = int(1 + 3.322 * np.log10(len(X)))

# Create Histogram bins
freq, bins = np.histogram(X, bins=k)

print("\nFrequency Distribution Table:")
for i in range(len(freq)):
    print(f"\t\t {round(bins[i])} - {round(bins[i+1])} : {freq[i]}")
```

```
Frequency Distribution Table:
                 11 - 16 : 8
                 16 - 20 : 9
                 20 - 24 : 9
                 24 - 29 : 5
                 29 - 34 : 6
                 34 - 38 : 3
```

d) Plots

In [11]:
```
def plots():
    plt.figure(figsize=(20, 5))
    plt.subplot(1, 4, 1)
    plt.plot(X)
    plt.title('Line Plot of X')

    plt.subplot(1, 4, 2)
    plt.hist(X)
    plt.title('Histogram of X')

    plt.subplot(1, 4, 3)
    plt.bar(range(len(X)), X)
    plt.title('Bar Plot of X')

    plt.subplot(1, 4, 4)
    plt.pie(X[:5], labels=['A','B','C','D','E'], autopct='%1.1f%%')
    plt.title('Pie Chart Example from X')

    plt.show()
```

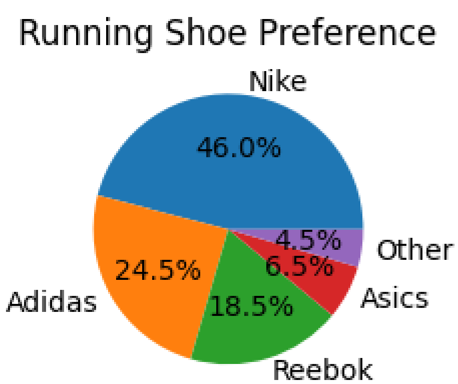
```python
plots()
```



Question 3

a) Pie chart of shoe preference

In [12]:
```python
brands = ['Nike','Adidas','Reebok','Asics','Other']
runners = [92,49,37,13,9]

plt.figure(figsize=(10, 52.5))
plt.subplot(1, 4, 1)
plt.pie(runners, labels=brands, autopct='%1.1f%%')
plt.title('Running Shoe Preference')
plt.show()
```



b) Histogram and Bar plot

In [13]:
```python
classes = ["0-10", "10-20", "20-30", "30-40", "40-50", "50-60"]
frequency = [8, 12, 30, 25, 18, 17]

plt.figure(figsize=(14,5))

# Bar Plot
plt.subplot(1,2,1)
plt.bar(classes, frequency)
plt.title("Bar Plot")
plt.xlabel("Class Interval")
plt.ylabel("Frequency")
```

```
plt.grid()

# Histogram
plt.subplot(1,2,2)
plt.bar(classes, frequency)
plt.title("Histogram")
plt.xlabel("Class Interval")
plt.ylabel("Frequency")
plt.grid()

plt.suptitle("Frequency Distribution Graphs")
plt.tight_layout()
plt.show()
```